

CE 1201

ENGINEERING MECHANICS

QUESTION BANK ANALYSIS

Set Distribution • Repeat Frequency • Theory • Proof/Derivation • Exam Strategy

9 Regular papers analysed	8 Study sets	3 of 4 / section Answer requirement
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Based strictly on the uploaded regular examination papers

2015 • 2016 • 2017 • 2018 • 2019 • 2021 • 2023 • 2024 • 2025

2020 is treated as an exception per the supplied note; no 2020 regular paper was present in the uploaded QB bundle, so it is not used in frequency counts. 2022 is also not present. Backlog 2023 is excluded from this regular-final analysis.

Made by 2223033

How to Use This Analysis

The report follows your set-distribution system, not the printed historical Section-A/Section-B labels, because the printed section labels changed across years.

Important interpretation Study Section A = Centroid / M.I. / Advanced Inertia / Dynamics. Study Section B = Force Systems / Beam / Truss / Friction-Cable cluster. Older and recent papers sometimes mix a companion subtopic inside a full question; those are flagged as cross-set spillovers.

- আপনাকে প্রতি study section থেকে ৪টি set-এর মধ্যে ৩টি answer করতে হবে। তাই preparation-এর লক্ষ্য শুধু “সব chapter একটু একটু” নয়; বরং ৩টি dependable set + ১টি insurance set তৈরি করা।
- Frequency = supplied regular papers-এর কত বছরে ওই exact question family এসেছে। 9/9 মানে সব ৯টি available regular paper-এ এসেছে।
- “Set repeat 9/9” মানে ওই set-এর anchor topic প্রতি বছর ছিল; কিন্তু subpart একই নাও হতে পারে।
- Most repeated items আগে সাজানো হয়েছে। শেষ মুহুর্তে Priority A items শেষ না করে low-frequency special problems-এ সময় দেবেন না।

Best 3-of-4 Strategy

Section	Core 3	Why	Insurance
Study Section A	A1 Centroid + A2 Area M.I. + A4 Dynamics	Most stable anchors.	A3 recent papers (2023-25) became stronger; learn Product of Inertia + standard mass-M.I. formulas as insurance.
Study Section B	B1 FBD/Equilibrium + B2 Beam + B3 Truss	Three 9/9 anchor families.	B4 is also frequent but exact friction/wedge/cable mix is less stable; at least prepare wedge + slide/tip + angle/repose.

Exam-smart rule একটি set পুরো বাদ দেওয়ার আগে অন্তত তার short theory + formula + ১টি most-repeated numerical pattern পড়ে রাখুন। কারণ full questions-এ cross-set subpart দেখা গেছে।

Overall Priority - Most Repeated Question Families

Freq.	Pattern	Rate	Set	Question family	Years
9/9		100%	A1	Composite centroid / C.G. of shaded or unshaded area	2015, 2016, 2017, 2018, 2019, 2021, 2023, 2024, 2025
9/9		100%	A2	Composite area moment of inertia (I_x , I_y / centroidal / parallel axis)	2015, 2016, 2017, 2018, 2019, 2021, 2023, 2024, 2025
9/9		100%	B2	Beam reactions + shear force + bending moment numerical	2015, 2016, 2017, 2018, 2019, 2021, 2023, 2024, 2025
9/9		100%	B3	Truss reactions/member-force numerical	2015, 2016, 2017, 2018, 2019, 2021, 2023, 2024, 2025

Freq.	Pattern	Rate	Set	Question family	Years
9/9		100%	B1	FBD / equilibrium modelling	2015, 2016, 2017, 2018, 2019, 2021, 2023, 2024, 2025
9/9		100%	A4	Dynamics full-question family (work/impulse/plane motion/impact)	2015, 2016, 2017, 2018, 2019, 2021, 2023, 2024, 2025
9/9		100%	B4	Friction/wedge/impending-motion family	2015, 2016, 2017, 2018, 2019, 2021, 2023, 2024, 2025
8/9		89%	B2	Define shear force and bending moment	2016, 2017, 2018, 2019, 2021, 2023, 2024, 2025
8/9		89%	A1	Centroid/C.G./C.O.M. theory question	2015, 2016, 2017, 2018, 2019, 2021, 2023, 2025
7/9		78%	A4	Linear impulse-momentum principle / proof	2015, 2016, 2017, 2018, 2021, 2023, 2024
7/9		78%	A2	Area M.I. definitions / concepts	2015, 2016, 2017, 2018, 2019, 2021, 2023
6/9		67%	B4	Angle of friction vs angle of repose	2015, 2016, 2019, 2021, 2024, 2025
6/9		67%	B4	Slide vs tip-over / overturning threshold	2015, 2016, 2017, 2018, 2021, 2025
6/9		67%	B2/B3	Pulley equilibrium numerical (appears in B2/B3)	2016, 2018, 2019, 2023, 2024, 2025
6/9		67%	B2/B4	Flexible/catenary/parabolic cable family	2015, 2017, 2021, 2023, 2024, 2025
5/9		56%	A3	Product of inertia of Z-section	2017, 2018, 2019, 2021, 2023
5/9		56%	B4	Wedge force / impending motion numerical	2016, 2019, 2023, 2024, 2025
4/9		44%	A2	$J = I_x + I_y$ derivation	2017, 2018, 2019, 2021
4/9		44%	A1	Centroid of circular sector	2017, 2019, 2023, 2025
4/9		44%	A4	Inclined-body work / kinetic-energy numerical	2018, 2019, 2021, 2023
4/9		44%	B2/B4	Cable span-sag / tension numerical	2017, 2023, 2024, 2025
4/9		44%	B4	Belt-friction derivation $T_1 = T_2 e^{(f\theta)}$	2015, 2019, 2021, 2024
3/9		33%	A3	Pappus-Guldinus application	2015, 2018, 2023
3/9		33%	A4	Motorcycle projectile / ditch problem	2015, 2018, 2021

Freq.	Pattern	Rate	Set	Question family	Years
3/9		33%	A4	Work against gravity - masonry chimney	2016, 2018, 2023
3/9		33%	A4/A3	Jet/blade impulse-momentum numerical	2015, 2016, 2025
3/9		33%	A1	Centroid of triangle derivation/location	2016, 2018, 2021

8-Set Distribution Overview

Set	Topic cluster	Anchor question	Repeat	Priority
A1	Centroid, Center of Gravity & Composite Areas	Composite centroid / C.G. numerical	9/9	ABSOLUTE MUST
A2	Moment of Inertia of Area, Polar M.I. & Radius of Gyration	Composite area I_x / I_y / centroidal or parallel-axis numerical	9/9	ABSOLUTE MUST
A3	Advanced Inertia: Product of Inertia, Mass M.I. & Pappus-Guldinus	Rotating/advanced inertia numerical (topic rotates by year)	9/9 as a set; exact subtopic varies	HIGH - VARIABLE SET
A4	Dynamics: Work-Energy, Impulse-Momentum, Plane Motion & Impact	Dynamics principle/proof + numerical	9/9	MUST - BROAD SET
B1	Force Systems, Equilibrium, FBD, Supports & Contact Reactions	Free-body diagram / equilibrium modelling	9/9	ABSOLUTE MUST
B2	Beam Reactions, Shear Force & Bending Moment + Cable/Pulley Companion	Beam reaction + S.F. & B.M. at specified section(s)	9/9	ABSOLUTE MUST
B3	Truss (Method of Joints/Sections) + Pulley/Plane-Motion Spillover	Truss reactions and member forces	9/9	ABSOLUTE MUST
B4	Friction, Sliding/Overtipping, Belt Friction, Wedge & Flexible Cord	Impending-motion friction / wedge / slide-vs-tip family	9/9 as a broad family	HIGH - VARIABLE SET

Year-wise Set Coverage Matrix

Year	A1	A2	A3	A4	B1	B2	B3	B4
2015	●	●	○	●	●	●	●	●
2016	●	●	○	●	●	●	●	●
2017	●	●	○	●	●	●	●	○
2018	●	●	●	●	●	●	○	●

Year	A1	A2	A3	A4	B1	B2	B3	B4
2019	●	●	●	●	●	●	●	⦿
2021	●	●	●	⦿	●	●	●	●
2023	●	●	●	●	●	●	●	⦿
2024	●	⦿	⦿	●	●	●	⦿	●
2025	●	⦿	●	⦿	●	●	⦿	●

Legend ● = clean anchor set; ⦿ = full question contains a significant companion/cross-set subtopic. This is why the safest preparation is 3 complete sets + a compact insurance layer for the 4th.

A1 — Centroid, Center of Gravity & Composite Areas

ANCHOR Composite centroid / C.G. numerical	REPEAT 9/9	PRIORITY ABSOLUTE MUST
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Bangla technique এটা সবচেয়ে নিরাপদ সেটগুলোর একটি। Composite shape কে basic shape-এ ভাগ করুন; hole/cut-out হলে area ঋণাত্মক নিন; reference axis আগে ঠিক করুন।

Year-wise Full-Question Pattern

Year	Question / subparts	Analysis note
2015	Q5: Define centroid; state Pappus-Guldinus; paint a cone-shaped monument; calculate $\bar{x}$ and $\bar{y}$ of a composite shaded area.	Pappus appears as a companion topic.
2016	Q5: Define centroid; real-life applications; locate centroid of a triangle (base b , altitude h); composite centroid numerical.	Classic centroid theory + composite math.
2017	Q5: Explain centroid and axis of symmetry; locate centroid of a circular sector; composite centroid numerical.	Sector formula is a repeat pattern.
2018	Q1: Define centroid; explain axis of symmetry; find centroid of a triangle; composite centroid numerical.	Very theory-friendly paper.
2019	Q1: Centroid vs center of gravity; applications; centroid of circular sector; first moment/composite centroid numerical.	Strong theory + math combination.
2021	Q5: Define centroid; applications; centroid of a triangle; composite centroid numerical.	Near-repeat of 2016.
2023	Q1: Distinguish centroid and C.G.; applications in building engineering; centroid of a circular sector; unshaded composite area centroid.	Latest-style wording.
2024	Q5: Centroid of area enclosed by parabolas $y^2 = 9x$ and $x^2 = 4y$; centroid of a composite unshaded area.	Special curve + composite numerical.

Year	Question / subparts	Analysis note
2025	Q5: Condition for center of mass to coincide with C.G.; centroid of a circular sector; composite C.G./centroid numerical.	Concept + sector + composite.

What to Prepare First

- 1) Composite centroid/C.G. numerical — 9/9.
- 2) Centroid/C.G./C.O.M. concepts — 8/9.
- 3) Circular sector centroid — 4/9.
- 4) Centroid applications — 4/9.
- 5) Triangle centroid — 3/9.
- 6) Axis of symmetry / special parabolic area — lower frequency but useful.

A2 — Moment of Inertia of Area, Polar M.I. & Radius of Gyration

ANCHOR Composite area I_x / I_y / centroidal or parallel-axis numerical	REPEAT 9/9	PRIORITY ABSOLUTE MUST
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Bangla technique প্রথমে centroid লাগবে কি না দেখুন। তারপর standard shape formula + parallel-axis theorem ব্যবহার করুন। Cut-out হলে M.I. বাদ যাবে। Unit সবসময় length^4 (যেমন mm^4 , in^4)।

Year-wise Full-Question Pattern

Year	Question / subparts	Analysis note
2015	Q6: Define moment of inertia and product of inertia; prove transfer/parallel-axis relation; compute I_x , I_y , centroidal M.I. and polar M.I. for a composite area.	Heavy formula + composite math.
2016	Q6: Define second moment and polar moment; radius-of-gyration/transfer relation; compute I_x and I_y of a shaded area.	Core MOI set.
2017	Q6: Define rectangular M.I., polar M.I. and radius of gyration; prove $J = I_x + I_y$; applications; composite M.I. about parallel/centroidal axes.	J-proof is highly reusable.
2018	Q2: Rectangular, polar and product of inertia; prove $J = I_x + I_y$; applications of first/second moment; composite second moment about x/y parallel and centroidal axes.	Full theory + numerical.
2019	Q2: Define rectangular/polar/product of inertia; prove $J = I_x + I_y$; calculate second moment of composite area about parallel and centroidal axes.	Near-repeat of 2018.
2021	Q6: Define M.I., polar M.I. and radius of gyration; prove $J = I_x + I_y$; composite M.I. about parallel and centroidal axes.	Very predictable.
2023	Q2: Define moment and product of inertia; triangular-area M.I./radius of gyration; composite polar M.I. and radius of gyration about parallel and centroidal axes.	Adds triangular derivation-style part.

Year	Question / subparts	Analysis note
2024	Q6 (dominant area-M.I. part): significance of M.I.; compute I_x and I_y of a composite shaded area. The same full question also contains a mass-M.I. derivation.	Cross-set mixing warning.
2025	Q6 (dominant area-M.I. part): compute moments of inertia about centroidal axes for a composite section; companion subpart is a work-energy train problem.	Cross-set mixing warning.

What to Prepare First

- 1) Composite I_x and I_y — 9/9.
- 2) Parallel-axis / centroidal-axis workflow — near-universal in numericals.
- 3) Definitions: rectangular/polar M.I., radius of gyration — very frequent.
- 4) $J = I_x + I_y$ proof — 4/9.
- 5) Polar M.I. + radius of gyration numerical — recurring advanced variant.
- 6) Applications/significance of M.I. — cheap theory marks.

A3 — Advanced Inertia: Product of Inertia, Mass M.I. & Pappus-Guldinus

ANCHOR Rotating/advanced inertia numerical (topic rotates by year)	REPEAT 9/9 as a set; exact subtopic varies	PRIORITY HIGH - VARIABLE SET
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Bangla technique এই সেট সবচেয়ে variable। শুধু একটি chapter পড়ে যাওয়া ঝুঁকিপূর্ণ। Minimum insurance: Product of Inertia (Z-section) + Mass M.I. standard formulas + Pappus দুই theorem।

Year-wise Full-Question Pattern

Year	Question / subparts	Analysis note
2015	Q7: Linear impulse/water-jet numerical + mass moment of inertia of a cast-iron frustum with cylindrical hole.	Legacy mixed question; mass M.I. is the A3 anchor.
2016	Q7: Work/power + $W_{\text{net}} = \Delta KE$ proof; chimney work; C.G. of a cone with cylindrical hole.	Legacy cross-set mixture.
2017	Q7: Shift of C.G. after making a square hole in circular lamina + product of inertia of a Z-section.	Product-of-inertia pattern begins.
2018	Q3: Pappus-Guldinus pulley-rim mass/weight + product of inertia of a Z-section.	Two classic A3 topics together.
2019	Q3: Applications of area M.I.; polar M.I./radius of gyration; product of inertia of a Z-section.	Z-section repeats again.
2021	Q7: Applications of area M.I.; polar M.I./radius of gyration; product of inertia of a Z-section.	Very close to 2019.
2023	Q3: Maximum/minimum M.I. from product of inertia; Pappus pulley-rim mass/weight; rotating disk-shaft angular deceleration.	Most comprehensive A3 paper.

Year	Question / subparts	Analysis note
2024	Q7: Mass M.I. of a frustum; impulse-momentum proof; work-energy inclined-body numerical.	Mass M.I. + dynamics mixed.
2025	Q7: Derive mass M.I. of a sphere and tangent-axis value; mallet radius of gyration; steam-jet force on blade.	Mass M.I. became prominent.

What to Prepare First

- 1) Product of inertia of Z-section — 5/9 (strongest exact A3 repeat).
- 2) Standard mass-M.I. formulas + parallel-axis theorem — increasingly important in 2023-25.
- 3) Pappus-Guldinus theorems + pulley-rim application — 3/9.
- 4) Maximum/minimum M.I. and principal values from P_{xy} — 2023 high-value special.
- 5) Radius of gyration of composite mass — recent trend.

A4 — Dynamics: Work-Energy, Impulse-Momentum, Plane Motion & Impact

ANCHOR Dynamics principle/proof + numerical	REPEAT 9/9	PRIORITY MUST - BROAD SET
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Bangla technique প্রশ্ন দেখেই method চিনুন: velocity-position → Work-Energy; velocity-time/short duration force → Impulse-Momentum; projectile → x-y motion আলাদা করুন। Free-body diagram ছাড়া dynamics math শুরু করবেন না।

Year-wise Full-Question Pattern

Year	Question / subparts	Analysis note
2015	Q8: Spring + friction work-energy problem; motorcycle projectile/ditch problem.	Two highly reusable numerical patterns.
2016	Q8: Linear impulse-momentum proof; impulse on a frictional block; moving-blade water-jet momentum problem.	Impulse-heavy set.
2017	Q8: Linear impulse-momentum proof; impulse of applied forces and net impulse on a frictional block.	Direct repeat of 2016 concept.
2018	Q4: Work and kinetic energy; inclined-body net work; impulse-momentum proof; work against gravity for chimney construction.	Mixed but very exam-friendly.
2019	Q4: Impulse/momentum definitions + work of system of forces; inclined-body work/KE; rolling-cylinder speed.	Work-energy dominant.
2021	Q4: Impulse-momentum proof; inclined-body work/KE; motorcycle projectile/ditch problem.	Combines 2017 + 2015 repeats.
2023	Q4: Classical dynamics; prove $W_{\text{net}} = \Delta KE$; impulse-momentum principles; work/KE terms; inclined-body net work; chimney work.	Strong theory + application.

Year	Question / subparts	Analysis note
2024	Q8: C.O.M. vs centroid condition; direct central impact of two freight cars; accelerated-car turn-over/relative-equilibrium problem.	Impact and non-inertial dynamics enter.
2025	Q8: Product-inertia applications + polar/radius numerical (spillover) + accelerated-car turn-over/relative-equilibrium problem.	Recent paper has significant inertia spillover.

What to Prepare First

- 1) Linear impulse-momentum principle/proof — 7/9.
- 2) Work-energy numerical on inclined body — 4/9.
- 3) Motorcycle projectile — 3/9.
- 4) Chimney work against gravity — 3/9.
- 5) Jet/blade momentum — 3/9.
- 6) Impact / accelerated-frame turn-over — recent 2024-25 trend.

B1 — Force Systems, Equilibrium, FBD, Supports & Contact Reactions

ANCHOR Free-body diagram / equilibrium modelling	REPEAT 9/9	PRIORITY ABSOLUTE MUST
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Bangla technique FBD-তে object isolate করুন → সব external force/reaction বসান → smooth surface হলে reaction normal → rope/cable tension একই line বরাবর। ভুল FBD হলে পুরো set নষ্ট হতে পারে।

Year-wise Full-Question Pattern

Year	Question / subparts	Analysis note
2015	Q1: Define/draw free-body diagram of members; wheel about to roll over a block - find least force.	FBD + wheel-over-block.
2016	Q1: Define transmissibility, two-force member, FBD, equilibrium, zero-force member; draw FBDs; resultant force system.	Maximum theory density.
2017	Q1: Define transmissibility, two-force member, equilibrium, FBD; draw FBD; wheel-over-block least force and reaction.	Classic repeat.
2018	Q5: Define transmissibility, two-force member, equilibrium, FBD; draw FBD; derrick cable tension and boom compression.	Two-force/boom equilibrium.
2019	Q5: Resultant force system; bell-crank reactions; draw FBD of a multi-member system.	Reaction + FBD emphasis.
2021	Q1: Define transmissibility, two-force member, FBD, equilibrium, zero-force member; draw FBDs; wheel-over-block least force.	Near-repeat of 2016/17.

Year	Question / subparts	Analysis note
2023	Q5: Define transmissibility, angle of repose, wedge, coplanar force system; draw FBD; support types + wheel/cylinder equilibrium with rope on incline.	Broader definition mix.
2024	Q1: Define concurrent force system, zero-force member, catenary chord; draw FBD; reactions at smooth contact surfaces of two bodies.	Contact-reaction variant.
2025	Q1: Define transmissibility, two-force member, FBD, rectilinear motion; draw FBD; smooth cylinder on incline with rope - find inclination/reaction.	FBD remains the anchor.

What to Prepare First

- 1) FBD drawing/equilibrium modelling — 9/9.
- 2) Transmissibility + two-force member + equilibrium definitions — repeated cluster.
- 3) Wheel-over-block least-force problem — 4/9 (2015, 2016, 2017, 2021).
- 4) Smooth-contact reactions — strong recent trend (2019, 2023, 2024, 2025).
- 5) Resultant force system — lower frequency but easy scoring.

B2 — Beam Reactions, Shear Force & Bending Moment + Cable/Pulley Companion

ANCHOR Beam reaction + S.F. & B.M. at specified section(s)	REPEAT 9/9	PRIORITY ABSOLUTE MUST
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Bangla technique Beam set-এর shortcut: (1) distributed load কে equivalent point load করুন, (2) support reactions বের করুন, (3) section cut, (4) left/right side দিয়ে shear ও moment cross-check করুন।

Year-wise Full-Question Pattern

Year	Question / subparts	Analysis note
2015	Q2: Catenary/parabolic-cord condition + prove catenary equation; beam reactions and S.F./B.M. at a section.	Cable proof + beam.
2016	Q2: Define S.F. and B.M.; simply-supported beam reactions and section values; pulley equilibrium to find W.	Beam + pulley.
2017	Q2: Define S.F./B.M. and find reactions; cable with span/sag - length, lowest-point tension and maximum tension.	Beam + cable numerical.
2018	Q6: Define S.F./B.M.; beam reactions + S.F./B.M. at two sections; pulley equilibrium.	Very standard.
2019	Q6: Define S.F./B.M.; beam reactions + S.F./B.M. at mid-span; frictionless pulley reaction/equilibrium.	Beam remains identical anchor.
2021	Q2: Conditions for parabolic/catenary chord + prove catenary equation; beam reactions + S.F./B.M. at two sections.	Near-repeat of 2015.

Year	Question / subparts	Analysis note
2023	Q6: Define torque/couple; beam reactions + S.F./B.M. at two sections; pulley equilibrium.	Companion theory changed.
2024	Q2: Define S.F./B.M.; beam reactions + S.F./B.M. at two sections; catenary cable length/tension/max tension.	Beam + cable.
2025	Q2: Define S.F./B.M.; beam reactions + S.F./B.M. at two sections; 60-ft cable with 360-lb weight and 15-ft sag - support tension/span.	Beam + cable remains strong.

What to Prepare First

- 1) Beam reactions + S.F./B.M. at section(s) — 9/9.
- 2) Define S.F. and B.M. — 8/9.
- 3) Cable/catenary/parabolic numerical or proof — 6/9 overall.
- 4) Pulley equilibrium — 6/9 overall (sometimes moves to B3).
- 5) Learn UDL/UVL equivalent loads and centroid locations cold.

B3 — Truss (Method of Joints/Sections) + Pulley/Plane-Motion Spillover

ANCHOR Truss reactions and member forces	REPEAT 9/9	PRIORITY ABSOLUTE MUST
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Bangla technique Truss-এ প্রথমে external reactions। তারপর zero-force member check। ৩টার বেশি unknown joint দিয়ে শুরু করবেন না। Specific members চাইলে Method of Sections দ্রুত। Tension/Compression sign শেষে স্পষ্ট লিখুন।

Year-wise Full-Question Pattern

Year	Question / subparts	Analysis note
2015	Q4: Define truss/couple/torque; truss assumptions; reactions and forces in selected members.	Theory + truss numerical.
2016	Q3: Inverted Howe roof truss - forces in members to the right of DE; companion wheel-over-block problem.	Truss is the dominant anchor.
2017	Q4: Forces in DE, EF and FH of a truss; companion wedge/plane-motion + boom equilibrium.	Method-of-section style.
2018	Q8: Motorcycle plane-motion numerical + forces in specified truss members.	Truss + plane motion.
2019	Q7: External reactions and loads/member forces of a truss; companion frictional two-block cable problem.	Truss + friction.
2021	Q8: Forces in DE, DG, EG, FG, FH of a truss; companion impulse problem.	Legacy mixed full question.
2023	Q7: External reactions and forces in GH, CG, GJ using method of sections; companion frictional block/cable problem.	Method of sections explicitly required.

Year	Question / subparts	Analysis note
2024	Q3: Belt-friction proof; truss forces in DE, EF, FH; pulley equilibrium.	Truss sits between proof + pulley.
2025	Q3: Parabolic-chord proof; truss reactions/member forces; pulley equilibrium.	Truss remains unavoidable.

What to Prepare First

- 1) Truss numerical — 9/9.
- 2) Method of Sections for selected members — especially 2016, 2017, 2023-25 style.
- 3) Method of Joints + zero-force member checks.
- 4) Pulley or friction companion subpart — recent tendency.
- 5) Plane motion can spill into this set in older papers.

B4 — Friction, Sliding/Overturning, Belt Friction, Wedge & Flexible Cord

ANCHOR Impending-motion friction / wedge / slide-vs-tip family	REPEAT 9/9 as a broad family	PRIORITY HIGH - VARIABLE SET
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Bangla technique Impending motion-এর direction আগে ঠিক করুন; friction সবসময় সম্ভাব্য motion-এর বিপরীতে। Slide vs tip-এ দুইটি threshold force আলাদা বের করে ছোটটি আগে ঘটবে। Wedge-এ contact reaction-এর direction খুব গুরুত্বপূর্ণ।

Year-wise Full-Question Pattern

Year	Question / subparts	Analysis note
2015	Q3: Angle of friction vs angle of repose; prove belt-friction relation; determine whether a tall slender body slides or tips.	A textbook friction set.
2016	Q4: Angle/repose; laws of friction; slide-vs-tip; wedge force Q (frictionless contacts).	Most complete friction set.
2017	Q3: Pulley equilibrium + slide-vs-tip of tall body. Q4 also contains wedge definition/plane motion as spillover.	Friction topic split across full questions.
2018	Q7: Slide-vs-tip; define wedge and plane motion; boom equilibrium.	Friction + wedge terminology.
2019	Q8: Prove belt-friction relation; wedge numerical; prove a constant-acceleration displacement equation.	Proof-heavy variant.
2021	Q3: Angle/repose; belt-friction proof; slide-vs-tip.	Near-repeat of 2015.
2023	Q8: Parabolic-chord derivation + cable tension/span; wedge force for impending motion.	Flexible cord + wedge.
2024	Q4: Angle of friction/repose; frictional two-body/pulley tension problem; wedge numerical.	Friction + wedge strongly represented.

Year	Question / subparts	Analysis note
2025	Q4: Angle of friction/repose; slide-vs-tip; wedge numerical.	Very predictable B4 pattern.

What to Prepare First

- 1) Angle of friction/repose + impending motion — very frequent.
- 2) Slide vs tip-over — 6/9.
- 3) Wedge numerical — 5/9.
- 4) Belt-friction proof — 4/9.
- 5) Two-body friction/pulley — 3/9.
- 6) Flexible/parabolic cord may spill into B4/B3; keep formulas ready.

Theory Question Bank — Priority Order

Memorise definitions in precise engineering language; use Bangla only to understand, but write the final exam answer in clean technical English unless instructed otherwise.

#	Theory cluster	Set	Priority	Exam note
1	Free Body Diagram (FBD), transmissibility, two-force member, equilibrium, zero-force member	B1	Very high	Definitions + drawing recur heavily.
2	Centroid, center of gravity, center of mass; differences/conditions/applications	A1	Very high	Often paired with sector/triangle/composite math.
3	Shear force, bending moment, moment, couple, torque	B2	Very high	S.F./B.M. definitions appear in 8/9 supplied papers.
4	Rectangular/polar/product M.I., radius of gyration, applications/significance	A2/A3	Very high	Theory often precedes a 20-25 mark numerical.
5	Angle of friction, angle of repose, friction laws, wedge	B4	High	Common short theory before large friction numerical.
6	Truss definition, assumptions, tension/compression, zero-force concept	B3	High	Numerical is universal; theory less frequent but cheap marks.
7	Work, power, kinetic energy, impulse, momentum, classical dynamics	A4	High	Repeated in theory/proof subparts.
8	Parabolic chord vs catenary chord; span, sag, tension terminology	B2/B4	High	Cable topic rotates between sets.
9	Product of inertia applications; maximum/minimum M.I.	A3	Medium-High	Especially important from 2017 onward and again in 2023/2025.

#	Theory cluster	Set	Priority	Exam note
10	Support types and reaction directions	B1	Medium	Directly asked in 2023; essential for FBD/SFBM anyway.

High-Value Short Definitions to Keep Ready

- Centroid; center of gravity; center of mass; axis of symmetry; axis of reference.
- Rectangular moment of inertia; polar moment of inertia; product of inertia; radius of gyration.
- Free body diagram; transmissibility; two-force member; zero-force member; equilibrium; concurrent/coplanar force system.
- Shear force; bending moment; moment of a force; couple; torque; support reaction.
- Truss; method of joints; method of sections; tension member; compression member.
- Static friction; kinetic friction; limiting friction; angle of friction; angle of repose; wedge.
- Parabolic chord; catenary chord; span; sag; maximum tension / lowest-point tension.
- Work; power; kinetic energy; impulse; momentum; classical dynamics; rectilinear motion.

Proof / Derivation Bank — Most Important First

#	Proof / derivation	Freq.	Years	Priority
1	Principle of linear impulse and momentum: $R\Delta t = \Delta(mv)$	7/9	2015, 2016, 2017, 2018, 2021, 2023, 2024	★★★★★
2	Polar-axis relation: $J = I_x + I_y$	4/9	2017, 2018, 2019, 2021	★★★★★
3	Belt friction: $T_1 = T_2 e^{\mu \theta}$	4/9	2015, 2019, 2021, 2024	★★★★★
4	Work-energy theorem: $W_{\text{net}} = \Delta KE$	2/9 explicit proofs	2016, 2023	★★★★★
5	Catenary equation: $y = k \cosh(x/k)$	2/9	2015, 2021	★★★★★
6	Parabolic chord relations: $y = wx^2/(2Q)$, $Q = wL^2/(8d)$	2/9 recent	2023, 2025	★★★★★
7	Mass M.I. derivation of standard solids (cylinder/sphere)	2/9 recent	2024, 2025	★★★★★
8	Parallel-axis / transfer formula for area M.I.	1-2 explicit papers	2015 (+ related 2016 form)	★★★
9	Centroid derivations: triangle / sector	Repeated as "find/locate"	2016, 2017, 2018, 2019, 2021, 2023, 2025	★★★★★
10	Constant-acceleration displacement relation	1/9	2019	★★

Core Equations You Should Recognise Immediately

Area centroid	$\bar{x} = \Sigma(A_i x_i) / \Sigma A_i$; $\bar{y} = \Sigma(A_i y_i) / \Sigma A_i$
Area parallel-axis theorem	$I = \bar{I} + Ad^2$
Polar moment	$J = I_x + I_y$

Product-of-inertia transfer	$P_{xy} = \bar{P}_{xy} + A d_x d_y$
Radius of gyration	$k = \sqrt{I/A}$ [area] ; $k = \sqrt{I/m}$ [mass]
Work-energy	$W_{\text{net}} = KE_2 - KE_1 = \Delta KE$
Linear impulse-momentum	$\int F dt = m v_2 - m v_1$; for constant resultant, $R\Delta t = \Delta(mv)$
Belt friction	$T_1 = T_2 e^{(f\theta)}$
Parabolic cable	$y = wx^2/(2Q)$; for equal-level supports, $Q = wL^2/(8d)$
Catenary	$y = k \cosh(x/k)$
Beam equilibrium	$\Sigma F_x = 0$; $\Sigma F_y = 0$; $\Sigma M = 0$

Numerical Problem Archetypes — What to Practise

Priority	Set	Archetype	Method checkpoint
★★★★★	A1	Composite centroid with triangle/rectangle/sector/arc/cut-out	Choose reference axes; split area; use negative area for holes; tabulate A, x, y, Ax, Ay.
★★★★★	A2	Composite I_x/I_y with holes + parallel/centroidal axes	Find centroid first if needed; use standard $\bar{I} + Ad^2$; subtract holes.
★★★★★	B2	Beam with point load + UDL/UVL + couple	Convert distributed loads; reactions; section cut; S.F./B.M. from both sides.
★★★★★	B3	Truss selected-member forces	Global reactions → method of sections/joints → label T/C.
★★★★★	B1	FBD + smooth contact / two-force members	Reaction direction first; then ΣF_x , ΣF_y , ΣM .
★★★★☆	B4	Slide vs tip-over	Calculate force for impending slide and impending tip separately; smaller threshold governs.
★★★★☆	B4	Wedge impending motion	Draw separate FBDs; place friction opposite impending relative motion at every contact.
★★★★☆	A3	Product of inertia of Z-section	Locate centroidal axes; use $\bar{P} + A d_x d_y$; respect quadrant signs.
★★★★☆	B2/B4	Cable span-sag-tension	Identify parabolic vs catenary loading; compute horizontal component then support tension/length.
★★★★☆	A4	Inclined body work-energy	Compute work of each force with sign; friction = μN ; apply $W_{\text{net}} = \Delta KE$.
★★★☆☆	A4	Impulse on block / jet on blade	Momentum components + impulse signs; be consistent with lb-slug-ft or SI units.

Priority	Set	Archetype	Method checkpoint
★★★★☆	A4	Motorcycle projectile over ditch	Resolve velocity; use horizontal travel for time, then vertical equation.
★★★★☆	A3	Mass M.I. of composite solids	Convert weight to mass if FPS; use standard solid M.I. + md^2 .
★★★★☆	B2/B3	Pulley equilibrium	Same rope → same tension; movable pulley load sharing; then beam moment equation if attached.

7-Step Final Preparation Plan

Step	Action	What exactly to do
1	Lock the guaranteed anchors	A1 composite centroid, A2 composite M.I., B1 FBD, B2 beam, B3 truss.
2	Finish the top proofs	Impulse-momentum, $J = I_x + I_y$, belt friction, work-energy, catenary/parabolic cable.
3	Build A3 insurance	One Z-section product-of-inertia problem + mass-M.I. formula sheet + one Pappus problem.
4	Build B4 insurance	One slide/tip + one wedge + angle/repose theory.
5	Do recent papers first	2025 t 2024 h 2023, en 2021, then older repeats.
6	Practise with timer	35 marks $\approx$ one full set. Practise selecting and finishing 3 sets per study section.
7	Make a 2-page memory sheet	Standard centroids, area/mass M.I., cable, friction, beam-load centroids, dynamics equations.

Recommended selection under pressure Section A: A1 + A2 + A4, with A3 insurance. Section B: B1 + B2 + B3, with B4 insurance. তবে question paper হাতে পাওয়ার পরে full question-এর সব subpart দেখে final 3-set selection করবেন—শুধু chapter name দেখে নয়।

What NOT to Do

- একটি formula দেখে numerical মুখস্থ করবেন না—figure/data বদলালেও method যেন একইভাবে করতে পারেন।
- FBD, centroid table, support reaction বা unit conversion skip করবেন না; এগুলোই সবচেয়ে বেশি cascading error তৈরি করে।
- A3/B4 পুরো বাদ দেবেন না; recent papers-এ cross-set mixing আছে।
- 2020/2022 নিয়ে frequency guess করবেন না—supplied regular QB source-এ paper নেই।

Methodology & Source Note

- Primary frequency base: uploaded CE 1201 Regular Final examination papers for 2015, 2016, 2017, 2018, 2019, 2021, 2023, 2024 and 2025.

- The uploaded set-wise notes were used to preserve the user-defined 8-set structure and topic grouping.
- The printed Section-A / Section-B labels in historical papers are not used as the stable study labels because the teacher/topic allocation changed across years.
- A “repeat” is counted by year, not by number of subparts. If the same family appears twice in one paper, that still counts as one year of occurrence.
- Cross-set mixed questions are explicitly flagged rather than silently forcing every subpart into one chapter.
- Backlog Examination 2023 is not included in the regular-exam frequency denominator.
- 2020 is marked as an exception according to the supplied instruction, but a 2020 regular paper is not present in the uploaded QB bundle; therefore no 2020 question wording or frequency was invented. 2022 is likewise absent.

Source Files Used

Type	File / group	Use in this report
Regular QB	Engineering Mechanics 15-21-merged.pdf	Contains supplied regular papers 2015-2019 and 2021.
Regular QB	CE1201 (2023).PDF	Regular Examination 2023.
Regular QB	CE 1201 QB 2024-25.pdf	Regular Examinations 2024 and 2025.
Set-wise A notes	Centroid / Moment of Inertia / Product of Inertia / Mass M.I. / Pappus / Dynamics / Impulse / Work notes	Used for set labels and topic vocabulary.
Set-wise B notes	Equilibrium/FBD / SFBM / Pulley / Truss / Plane Motion / Friction / Flexible Cord notes	Used for set labels and topic vocabulary.

END OF ANALYSIS

Made by 2223033